

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element A Initial Template

How to use this template

Students: The details below are commonly found in problem definitions, but the specific assignment given by your instructor may dictate different details. If you make a copy of this template to edit for your own problem definition, be sure to delete the instructions.

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Explain your problem in detail, including details that you might consider common knowledge. Use data and statistics to justify your problem statement. Do not provide a bulleted list of facts; rather, organize the problem details in paragraph form. Address as many of the 5 Ws +1 H (who, what, where, when, why, how) as is possible, for example:

- **A1. Problem Identification:** A new construction apartment building is in need of building an alarm system that notifies residents of a fire and successfully puts it out. Currently, there is no way to detect a fire if one was started and no successful way to put it out to ensure that it doesn't spread. Without the creation of this system, it lacks safety in the building and for the residents if a situation were to arise.
- **A2. Problem Definition:** Two girls who share a high end apartment decided to have a girls night on Friday, February 6th. They baked cookies and then sat in the living room to watch movies together. While the girls are watching the movie they smell something burning.
- Where is the problem happening?
 - ◆ Inside of the apartments in the apartment buildings.
- When does the problem occur?
 - ◆ Can occur at any time when plausible factors to start a fire are present.
- Why does the problem happen?
 - ◆ Mostly human error, complete accident, electrical errors.
- How often does the problem happen?

Name: _____

Date: _____

Class: _____

- ◆ In the U.S., home fires happen every 88-89 seconds. Over 350,000 to nearly 500,000 home-based or residential structure fires are reported annually. (The Zebra, Jan. 15th, 2026).

In order to earn a score of 5, you will need to cite at least five sources that are from the last ten years and that come from 3 or more of the following credible categories: peer reviewed journals, professional organizations, government agencies, local and national newspapers. Use in-line citations to explicitly state which information is from which source. Do not just list all of the citations at the end of the submission or in footnotes.

Stakeholder Concerns and Connections (A3, A4)

Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem. Consider the breadth of the problem and how it might have social and environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system.

A3. Problem Elaboration If there is no alarm system in the apartments, fires would spread rapidly without residents being notified and there would be no plausible way to put it out. Long-lasting effects of fire could reside with residents like irritation, respiratory distress, cardiovascular issues, and more.

A4. Concern of Primary Stakeholders Groups

- **Residents:** Directly affected because they'll need an early alert system and safe evacuation routes in order to exit safely and quickly with effective equipment. The need to be confident that the alarm will work and provide an early warning and that the evacuation routes are clear and safe.
- **The property owner/Landlord:** Will be held accountable for the fire. They're concerned with ensuring that the system is reliable and compliant with the buildings codes and safety regulations.
- **Installers/Developers:** Their concern is around the technical aspects of the project. They need to ensure that it is a proper design, integrates with other building systems, and defect free.

Name: _____

Date: _____

Class: _____

- **First Responders:** Going to regulate the situation and help residents be removed from the home and area in order to eliminate any further issues or deaths. Their primary concern is to arrive in a timely manner and have accurate information about the fire's location to have a swift and effective response. Minimizes the risks to the residents and firefighters.
- **Insurance companies:** Will put money into repairs and cover any damages to oneself and property. Their main concern is to reduce financial risk. An effective fire alert system makes the likelihood of more property damage and injuries less.

Problem Importance (A6)

End Element A with a compelling argument for why this problem is important and needs to be solved now. Support this argument with data and statistics.

Fire safety in apartment buildings is crucial because residential fires happen often and can quickly lead to deaths, injuries, and significant property loss. Fire alarm systems provide early warnings so people escape safely, and fire suppression systems like sprinklers help control or extinguish fires before they spread. Together, these systems greatly reduce risk and protect both lives and property. According to

<https://www.pew.org/en/research-and-analysis/issue-briefs/2025/09/modern-multifamily-buildings-provide-the-most-fire-protection> and statistics shown modern family homes were deemed safer than single family homes and less than one sixth of the death rate in other housing types.

Sources Used:

<https://www.safefire.org/resources/fire-prevention-guide/>

<https://www.pew.org/en/research-and-analysis/issue-briefs/2025/09/modern-multifamily-buildings-provide-the-most-fire-protection>

<https://nfpa.org/2025/07/29/maryland-leads-by-example-what-16-years-of-fire-data-and-20-years-of-national-research-say-about-residential-automatic-fire-sprinklers/>

<https://candofp.us/blog/2025/03/11/fire-sprinkler-inspections/>

<https://ctif.org/index.php/news/use-automatic-fire-sprinklers-domestic-environments>

